Chester Grudzinski

Fort Worth, TX 76107

817-907-9965 | cwgrudzinski@gmail.com | linkedin.com/in/chester-grudzinski

https://github.com/generlmoo/

Professional Summary

Software Engineer with experience building backend systems, data ingestion pipelines, automation platforms, and cloud-based applications using TypeScript, Node.js, PostgreSQL, Python, and AWS. Experienced in production reliability, security remediation, cron orchestration, and large-scale data processing for financial analytics systems.

Work Experience

TickerTrends | *Full Stack Software Engineer - Financial Analytics Platform* **Jan 2026 - Present**

- Full-stack engineer on an alternative-data financial analytics platform built with TypeScript, PostgreSQL, AWS, Redis, Docker, and React, developing forecasting systems, data pipelines, and reporting infrastructure used for equity research and market intelligence.
- Built a production report platform for delivering financial trend metrics using Playwright rendering, async job orchestration, WebSocket progress streaming, S3 caching, block-based templates, and LLM-powered summary generation.
- Hardened production security and performance by patching ~70 SQL-injection vulnerabilities across 27 service files, securing internal endpoints, optimizing YoY computation from $O(n^2)$ to $O(n)$, and preventing Chromium OOM failures through concurrency limits.
- Re-architected cron orchestration by migrating tracking from Redis to PostgreSQL, adding health and error metadata across 39 scheduled workloads, and isolating cron processing onto dedicated backend infrastructure to improve reliability and observability.
- Built resilient ingestion and analytics pipelines processing alternative datasets from social, search, web, and RSS sources, handling Cloudflare-blocked responses, malformed feeds, rate-limited APIs, and multi-year historical backfills.

Young and Associates Inc. | *IT Support Specialist - Accounting Firm* **Jul 2025 - Jan 2026**

- Designed and deployed FastParse, a Python-based GUI tool that parses bank-issued PDF statements into structured accounting data and imports transactions directly into QuickBooks, eliminating manual preprocessing and reducing reliance on paid third-party applications.
- Automated depreciation and amortization workflows by developing a Python automation bot for ATX, increasing data-entry throughput from 2.7 to 12.5 accounts per minute and reducing manual errors.
- Recovered critical financial data from a compromised BitLocker-encrypted NVMe drive, performing forensic-safe imaging, decryption analysis, and full Windows system rebuild to restore business continuity.

Dallas Formula Racing | *Systems Engineer - Battery Management System* **May 2024 - Dec 2024**

- Designed and assembled a 720 V high-voltage battery management system (BMS) in Altium Designer; produced a functional prototype that passed all electrical safety tests.
- Developed embedded firmware for Texas Instruments battery-management hardware to monitor voltage, current, and I^2C-connected temperature sensors (TMP100) for real-time thermal monitoring and early fault detection; collaborated on thermal simulation that reduced peak cell temperature during simulated load testing.

Education

University of Texas at Dallas **Jan 2021 - Dec 2025**
Bachelor's degree, Computer Engineering (ABET Accredited) *Erik Jonsson School of Engineering and Computer Science*

Tarrant County College **Jan 2018 - Dec 2020**
Associate's degree, Information Technology

Projects

Raspberry Pi Self-Hosted NAS | *generlmoo.me* **Sep 2025 - Present**

- Designed and deployed a self-hosted Raspberry Pi 5 infrastructure platform featuring NVMe-backed storage, Cloudflare Tunnel remote access, OpenMediaVault monitoring, FileBrowser administration, automated backups, and Linux-based service orchestration.

Temperature Control System | *Dallas Formula Racing* **Apr 2024 - Dec 2024**

- Implemented the I^2C protocol in C (Arduino) to read temperature data from a TI TMP100 sensor using an ESP32-C3, including hand-soldering and correct I^2C pull-up resistor design.
- Developed an embedded Wi-Fi HTTP server to display real-time temperature data for remote monitoring.

Skills

- **Programming Languages:** Java, Python, JavaScript, TypeScript, SQL, C
- **Infrastructure:** Linux (Debian), AWS, PostgreSQL, Node.js, TypeORM, WebSockets, REST APIs, Redis, React, Next.js
- **Embedded & Hardware:** Embedded Systems (C), SPI, I^2C, UART, Soldering, Oscilloscopes, PCB Design
- **Tools:** Git, VS Code, MATLAB, Verilog, Arduino IDE, Altium Designer, Multimeters, KiCAD